Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

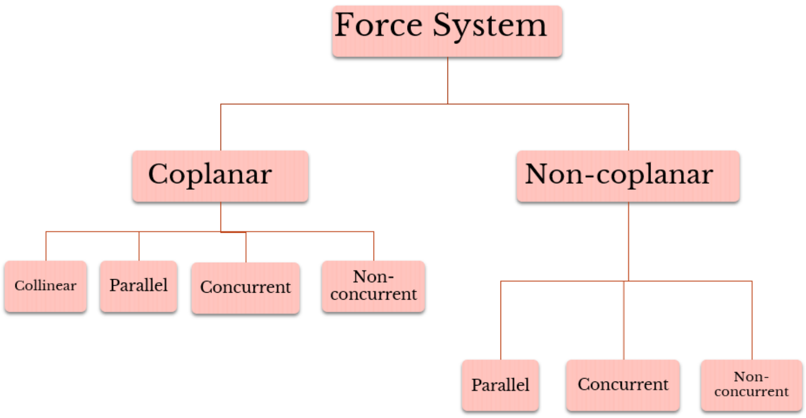
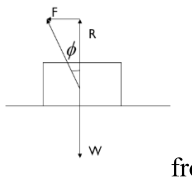
FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021

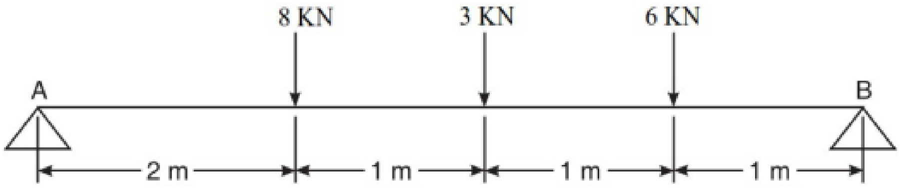
Course Code: EST100

Course Name: ENGINEERING MECHANICS

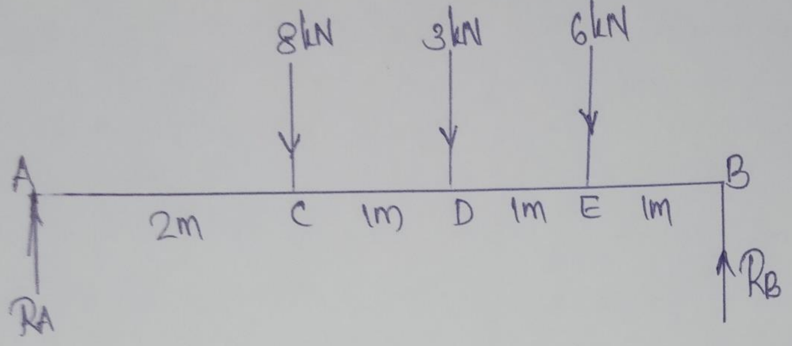
PART A

Answer all questions. Each question carries 3 marks

Q1	List out and explain systems of forces.
Ans	 When all forces are acting in the same plane, they are called coplanar force system. When the forces are acting in different planes is called non coplanar force system.
Q2	State & Explain the Varignon's theorem
Ans	Varignon's theorem states that the moment of the resultant about any point is equal to the algebraic sum of moments of all forces of the system about the same point. The moment of a force about any point is equal to algebraic sum of moments of its components about that point.
Q3	Define coefficient of friction. Show that the coefficient of friction is tangent of the angle of friction
Ans	Coefficient of friction (μ) – Ratio of limiting force of friction (*F*) to the Normal Reaction (R) $\mu = \frac{F}{R}$ Angle of friction (φ)-The angle which the resultant of the normal reaction and limiting force of friction makes with the normal reaction.  from the right-angled triangle, Tan $\varphi = \frac{F}{R} = \mu$

	Q4	Find the reactions at A and B 

Ans



$\underline{\Sigma f_y = 0}$

$R_A - 8 - 3 - 6 + R_B = 0$

$R_A + R_B = 17\,kN$ ———— (1)

$\underline{\Sigma M_A = 0}$

$-(8\times2)-(3\times3)-(6\times4)+(R_B\times5) = 0$

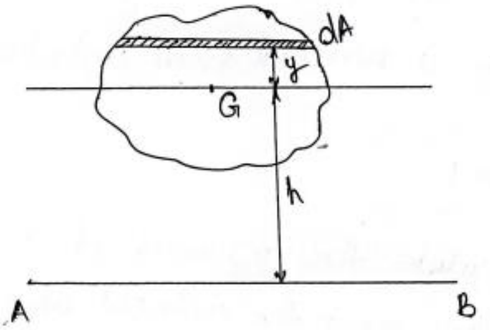
$5R_B = 16+9+24$

$R_B = \frac{49}{5} = \underline{\underline{9.8\,kN}}$

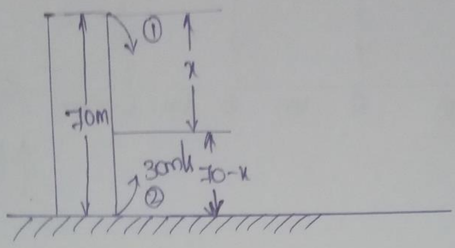
Substitute in eq (1)

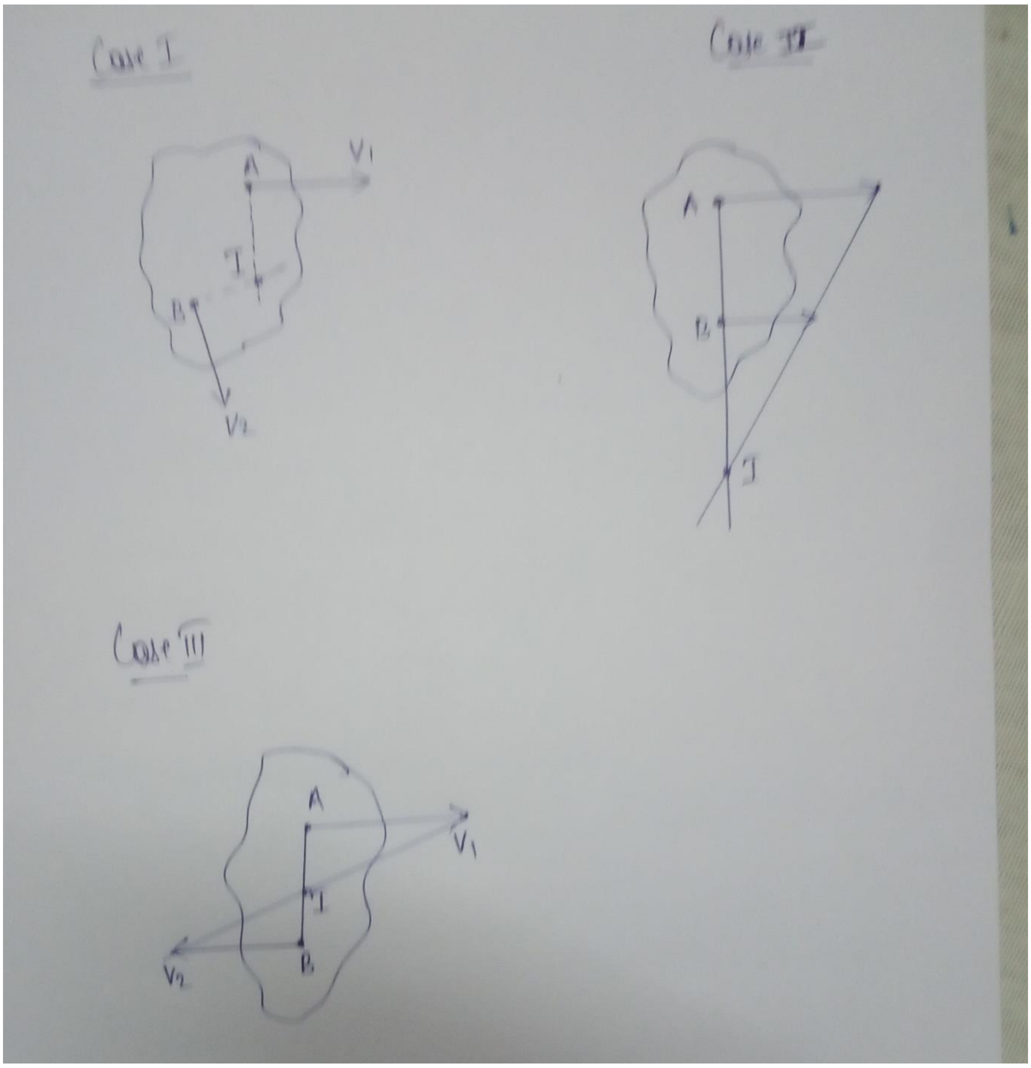
$\Rightarrow R_A + 9.8 = 17$

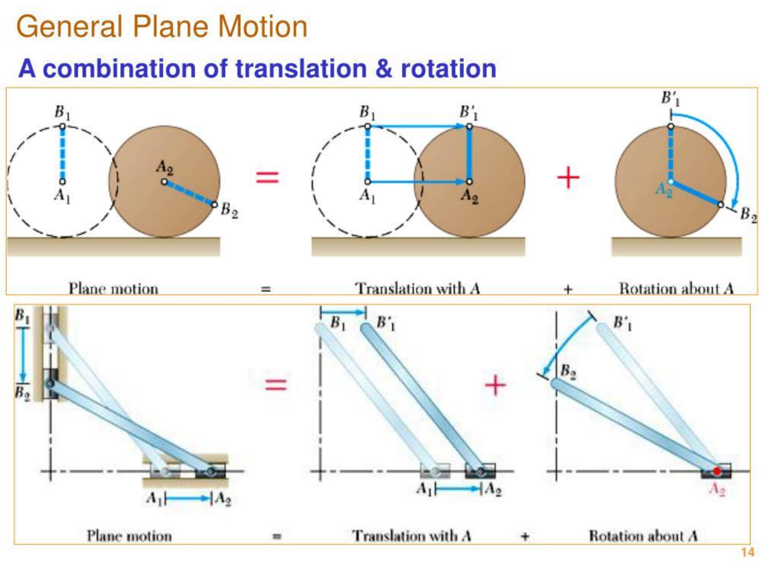
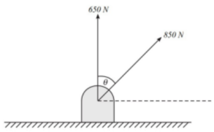
$R_A = \underline{\underline{7.2\,kN}}$

	Q5	Discuss the generation of area by theorem of Pappus Guldinus
	Ans	The area of surface generated by revolving a plane curve about a non-intersecting axis in the plane of the curve is equal to the product of length of curve and the distance travelled by the centroid ‘G’ of the curve during revolution. Area generated = $2\pi\hat{y}L$
	Q6	State and explain parallel axis theorem.
	Ans	It states that if IGG is the moment of inertia of a plane lamina of area A, about its centroidal axis of the lamina, then the moment of inertia about any axis AB which is parallel to the centroidal axis and at a distance h from the centroidal axis is given by $$I_{AB} = I_{GG} + Ah^2$$  Consider an elemental area dA, at a distance y from centroidal axis. The second moment of area about axis AB is $\int dA \cdot (y+h)^2$ $I_{AB} = \int dA(y+h)^2 = \int dA(y^2+h^2+2yh)$ $= \int dAy^2 + \int dA h^2 + \int dA \cdot 2yh$ We know, $\int dAy^2 = I_G$. $\int dA \cdot h^2 = h^2 \cdot \int dA = h^2 \cdot A$. $\int dA \cdot 2yh = 2h\int y dA = 2h \cdot A\bar{y} = 2h \cdot A \times 0 = 0$

		$\therefore I_{AB} = I_G + Ah^2 + 0$ $I_{AB} = I_G + Ah^2$. $\bar{y} = 0$, because it is the distance of centroid G from the axis from which y is measured. In the figure y is measured from centroidal axis itself.
	Q7	Discuss the use of D-Alembert's principle used for the analysis of a moving rigid body
	Ans	The system of forces acting on a moving body is in dynamic equilibrium with the inertia force of the body. It is also known as equation of dynamic equilibrium. $F – ma = 0$
	Q8	A stone is dropped from the top of a tower 70 m high. At the same time another stone is thrown up from the foot of the tower with a velocity of 30 m/s. At what distance from the top and after how much time the two stones cross each other?

	Ans	 Stone 1 $u_1 = 0$ $a_1 = g$ $t_1 = t$ $s_1 = x$ $S = ut + \frac{1}{2}at^2$ $x = 0 + \frac{1}{2} \times 9.81 \times t^2$ ——(1) Stone 2 $u_2 = 30m/s$ $a_2 = -g$ $t_2 = t$ $S_2 = 70 - x$ $S = ut + \frac{1}{2}at^2$ $70 - x = 30t + \frac{1}{2} \times -9.81 \times t^2$ ——(2) $70 - \frac{9.81t^2}{2} = 30t - \frac{9.81}{2}t^2$ $30t = 70$ $t = 2.33 sec$ Substitute $t = 2.33$ in eq(1) $\Rightarrow x = \frac{9.81}{2} \times (2.33)^2$ $= 26.70m$
	Q9	What do you mean by instantaneous centre of rotation? How can it be located for a body moving with combined motion of rotation and translation?

	Ans	The motion of rotation and translation of a body at a given instant, can be considered as that of pure rotation of the body about a point. This point about which the body can be assumed to be rotating at the given instant is called instantaneous centre of rotation. 
	Q10	What do you mean by general plane motion? Give two examples of bodies performing combined motion of rotation and translation
	Ans	A body is said to have general plane motion, if it possesses translation and rotation simultaneously. Common example of such motion is a wheel rolling on straight line.

		
PART B		
Answer any one full question from each module. Each question carries 14 marks		
Module 1		
	Q11 a	Determine angle between the forces and the direction of the resultant shown in figure. The resultant of the two forces is 1300 N. 
	Ans	

Given:-

$P = 650N$

$Q = 850N$

$R = 1300N$

We know, $R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$

where, $\theta \rightarrow$ Angle between forces P and Q

ie $1300 = \sqrt{650^2 + 850^2 + 2\times650\times850\times\cos\theta}$

$1300^2 = 650^2 + 850^2 + 2\times650\times850\times\cos\theta$

$\cos\theta = \dfrac{1300^2 - 650^2 - 850^2}{2\times650\times850} = 0.4932$

$\theta = \cos^{-1}(0.4932) = \underline{\underline{60.44^\circ}}$

Direction of resultant,

$\tan\beta = \dfrac{Q\sin\theta}{P + Q\cos\theta}$

where, $\beta \rightarrow$ Angle between resultant and P

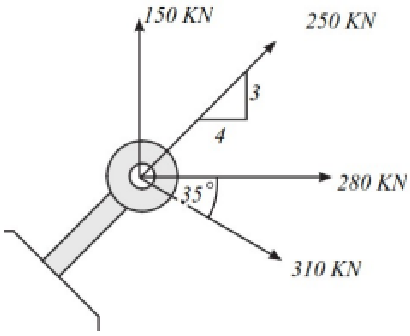
$\theta \rightarrow$ Angle between P and Q

$\tan\beta = \dfrac{850\sin 60.44}{650 + 850\cos 60.44}$

$= 0.6914$

$\beta = \tan^{-1}(0.6914)$

$= \underline{\underline{34.66^\circ}}$

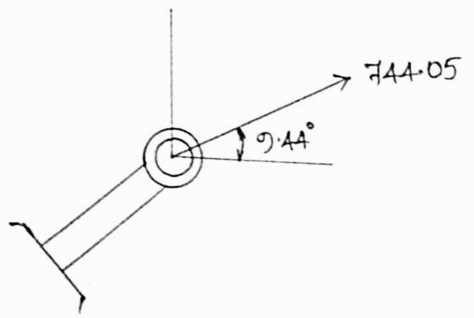
<table>
<tr><td></td><td>Q11
b</td><td>Four forces are acting on a bolt as shown in Figure 3.10. Determine the magnitude and direction of the resultant force

</td></tr>
<tr><td></td><td>Ans</td><td>In order to get the angle between 250kN and horizontal axis, consider the right angled triangle
$\tan\theta = \frac{3}{4}$
$\theta = \tan^{-1}(3/4) = 36.86^{\circ}$
Now in order to find the resultant,
Magnitude, $R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$
Direction, $\tan\theta = \frac{\Sigma F_y}{\Sigma F_x}$</td></tr>
</table>

Force	F_x	F_y
150kN	0	+150
250kN	+250 Cos 36·86	+250 Sin 36·86
280kN	+280	0
310kN	+310 Cos 35	-310 Sin 35

$\Sigma F_x = +733{\cdot}96 \qquad \Sigma F_y = +122{\cdot}15$

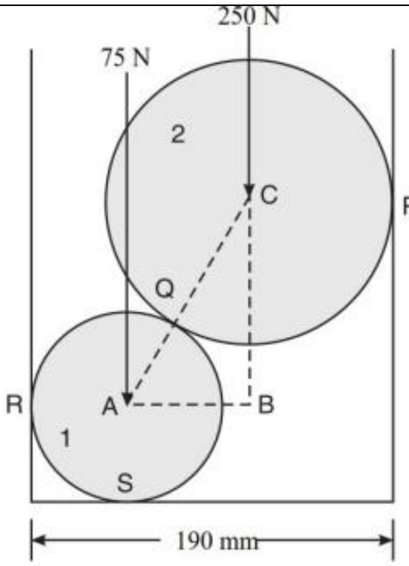
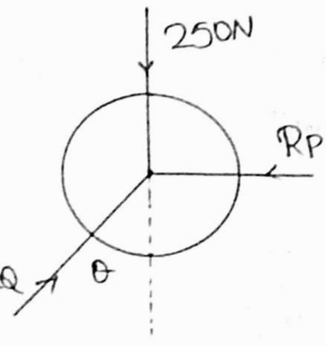
Resultant, $R = \sqrt{733{\cdot}96^2 + 122{\cdot}15^2} = \underline{\underline{744{\cdot}05\,kN}}$

$\tan\theta = \dfrac{\Sigma F_y}{\Sigma F_x} = \dfrac{122{\cdot}15}{733{\cdot}96} \Rightarrow \theta = \underline{\underline{9{\cdot}44^\circ}}$

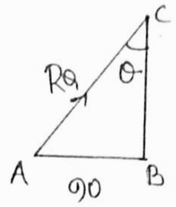


Module 2

Q12	Determine the reactions at contact points P, Q, R, and S for the system shown in Figure. The radii of spheres 1 and 2 are, respectively, 40 mm and 60 mm.

		
	Ans	Consider Free Body Diagram of sphere 2, 

To get the angle between R_Q with vertical, consider right angled triangle ABC



AC = Radius of 1 + Radius of 2
= 40 + 60
= 100 mm

AB = 190 − 40 − 60
= 90 mm

$$\sin\theta = \frac{90}{100}$$

$$\theta = \sin^{-1}(9/10)$$

$$\theta = 64.15^\circ$$

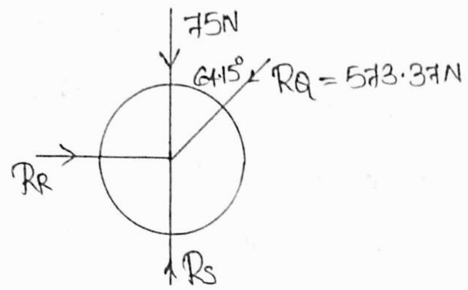
Now, Apply Lami's theorem,

$$\frac{250}{\sin 154.15} = \frac{R_P}{\sin 115.85} = \frac{R_Q}{\sin 90}$$

$$R_P = \frac{250 \times \sin 115.85}{\sin 154.15} = \underline{\underline{516\,N}}$$

$$R_Q = \frac{250 \times \sin 90}{\sin 154.15} = \underline{\underline{573.37\,N}}$$

Now consider FBD of sphere 1



Apply conditions of equilibrium, $\Sigma F_x = 0$, $\Sigma F_y = 0$

Force	F_x	F_y
75	0	-75
R_R	$+R_R$	0
R_S	0	$+R_S$
573.37	$-573.37 \sin 64.15$	$-573.37 \cos 64.15$

$\Sigma F_x = 0$

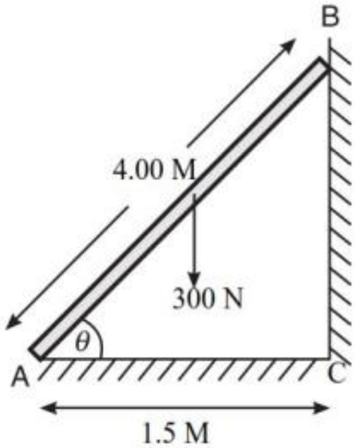
$R_R - 573.37 \sin 64.15 = 0$

$R_R = \underline{\underline{516N}}$

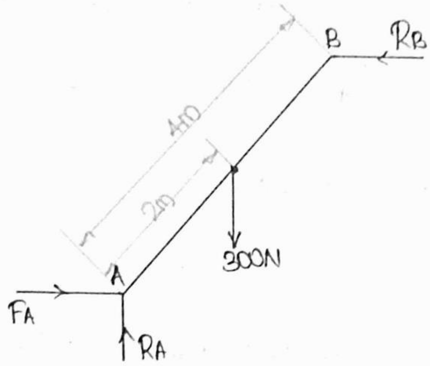
$\Sigma F_y = 0$

$-75 + R_S - 573.37 \cos 64.15 = 0$

$R_S = \underline{\underline{325N}}$

	13 a	A uniform ladder AB of length 4.00 m and weighing 300 N is placed against a smooth wall with its lower end 1.50 m from the wall. The coefficient of friction between the ladder and the floor is 0.25. What is the frictional force acting at the point of contact between the ladder and the floor? Show that the ladder will remain in equilibrium in this position. 
	Ans	Given:- $AB = 4m$ $AC = 1.5m$ $W = 300N$ $\mu_A = 0.25$ $\cos\theta = \frac{AC}{AB} = \frac{1.5}{4}$ $\theta = \cos^{-1}(0.375)$ $\theta = \underline{67.97^\circ}$

Consider FBD of ladder,



Now, Apply the conditions of equilibrium,

$\underline{\Sigma F_x} = 0$

$F_A - R_B = 0$

$F_A = R_B$ ——— (1)

$\underline{\Sigma F_y} = 0$

$R_A - 300 = 0$

$R_A = 300N$

Substitute in eq(1)

$F_A = R_B$

$0.25 R_A = R_B$

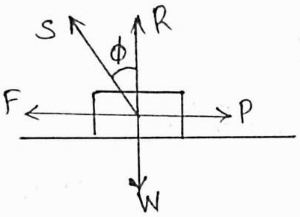
$0.25 \times 300 = R_B = \underline{\underline{75N}}$ $\quad$ $F_A = \underline{\underline{75N}}$

		To show that the ladder will remain in equilibrium in this position, take the moment with respect to point A, $-(300 \times 2 \cos 67.07) + (R_B \times 4 \sin 67.07) = 0$ $R_B = \underline{60.69N}$ $F_A = \underline{60.69N}$ And now we already calculated $F_A = 75N > 60.69$
	13 b	Explain angle of friction and angle of repose. Show that angle of repose is equal to angle of friction.

Ans

Angle of friction

The angle which the resultant of normal reaction and limiting force of friction makes with normal reaction.



Where,

$W \rightarrow$ Weight of body

$P \rightarrow$ Applied force

$R \rightarrow$ Normal reaction

$F \rightarrow$ Limiting frictional force

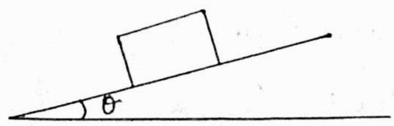
$S \rightarrow$ Resultant of normal reaction & limiting frictional force

From the figure,

$$\tan\phi = \frac{F}{R} = \mu \quad \text{——} (1)$$

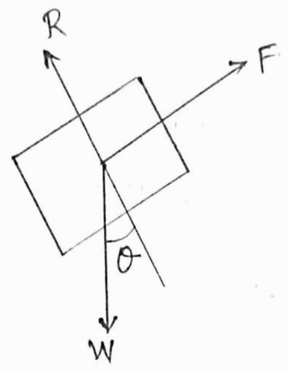
Angle of repose

It is the maximum inclination of a plane on which a body can repose without applying external force.



$\theta \rightarrow$ Angle of repose

Consider FBD of block,



Apply conditions of equilibrium,

$\Sigma F \perp r$ to plane $= 0$

~~$R - W \sin\theta = 0$~~

~~$R = W \sin\theta$~~

$R - W \cos\theta = 0$

$R = W \cos\theta$ ——(a)

ΣF parallel to plane $= 0$

~~$F - W \cos\theta =$~~

$F - W \sin\theta = 0$

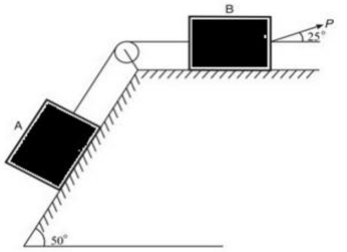
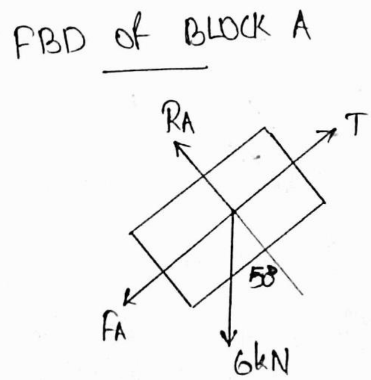
$F = W \sin\theta$ ——(b)

$$\frac{(b)}{(a)} = \frac{F}{R} = \frac{W \sin\theta}{W \cos\theta} = \tan\theta$$

$$\tan\theta = \frac{F}{R} = \mu \quad \text{——(2)}$$

Compare equations (1) and (2)

$$\tan\phi = \tan\theta = \frac{F}{R} \Rightarrow \underline{\underline{\theta = \phi}}$$

	14	Two blocks A and B weighing 6 kN and 3.5 kN, respectively, are connected by a wire passing over a smooth frictionless pulley as shown in Figure. Determine the magnitude of force P which is applied on block B at 25o from horizontal as shown in figure. Take μ = 0.20. 
	Ans	

ΣF perpendicular to plane $= 0$

$$R_A - 6 \cos 50 = 0$$

$$R_A = 6 \times \cos 50$$

$$= \underline{\underline{3.85 \text{kN}}}$$

ΣF parallel to plane $= 0$

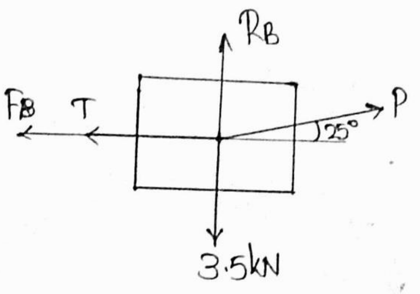
$$T - F_A - 6 \sin 50 = 0$$

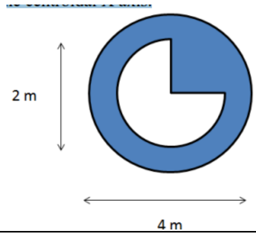
$$T - \mu R_A - 6 \sin 50 = 0$$

$$T = 6 \sin 50 + 0.2 \times 3.85$$

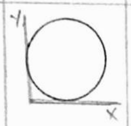
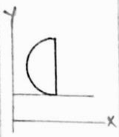
$$= \underline{\underline{5.36 \text{kN}}}$$

Now consider FBD of BLOCK B



		Apply conditions of equilibrium, $\sum F_y = 0$ $R_B + P \sin 25 - 3.5 = 0$ $R_B = 3.5 - P \sin 25$ —— (a) $\sum F_x = 0$ $P \cos 25 - T - F_B = 0$ $P \cos 25 - 5.36 - 0.2 R_B = 0$ $P \cos 25 - 5.36 - 0.2(3.5 - P \sin 25) = 0$ $P \cos 25 - 5.36 - 0.7 + 0.2 P \sin 25 = 0$ $P(\cos 25 + 0.2 \sin 25) = 5.36 + 0.7$ $P = \frac{6.06}{0.99}$ $P = \underline{6.12 kN}$
	15	From a circular lamina of diameter 4m, 3/4th quarter circle of diameter 2m has been remove from the centre. Determine the moment of inertia of the resulting composite figure about the centroidal X axis. 

	Ans	
		To locate the centroid of the shaded area, split the total area into three parts Area1 – Circular lamina of diameter 4m Area2 – Semicircle with radius 1m Area 3 – Quarter circle with radius 1m Shaded Area = Area 1 – Area 2 – Area 3 Now consider bottom most line as reference x axis and leftmost line as reference y axis.

SL No.	Figure	Area (A_i)	x_i	y_i
1.		πr^2 $\pi \times 2^2$ $= 4\pi$	2	2
2.		$\pi r^2/2$ $= \pi \times 1^2/2$ $= 0.5\pi$	$2 - 4r/3\pi$ $2 - \frac{4\times1}{3\pi}$ $= 1.57$	2
3.		$\pi r^2/4$ $= \pi \times 1^2/4$ $= 0.25\pi$	$2 + 4r/3\pi$ $2 + \frac{4\times1}{3\pi}$ $= 2.42$	$2 - \frac{4r}{3\pi}$ $2 - \frac{4\times1}{3\pi}$ $= 1.57$

$$\bar{x} = \frac{A_1x_1 - A_2x_2 - A_3x_3}{A_1 - A_2 - A_3}$$

$$= \frac{4\pi\times2 - 0.5\pi\times1.57 - 0.25\pi\times2.42}{4\pi - 0.5\pi - 0.25\pi}$$

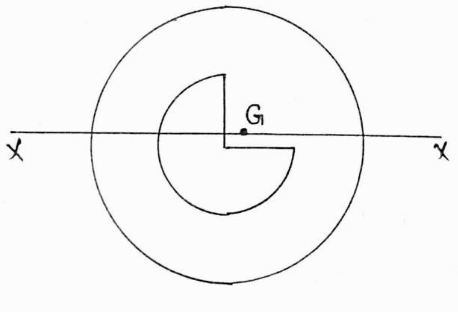
$$= \frac{\pi(8 - 0.785 - 0.605)}{\pi(3.25)} = \underline{\underline{2.033m}}$$

$$\bar{Y} = \frac{A_1 y_1 - A_2 y_2 - A_3 y_3}{A_1 - A_2 - A_3}$$

$$= \frac{4\pi \times 2 - 0.5\pi \times 2 - 0.25\pi \times 1.57}{4\pi - 0.5\pi - 0.25\pi}$$

$$= \frac{\pi(8 - 1 - 0.3925)}{\pi(3.25)} = \underline{\underline{2.033\,m}}$$

Next step is to find the moment of inertia with respect to centroidal axis.



SL No.	Figure	Area	I_{Gxx}	dy	$A(dy)^2$
1.		4π	$\frac{\pi D^4}{64}$ $= \frac{\pi \times 4^4}{64}$ $= 4\pi$	$y_1 - \bar{y}$ $= 2 - 2.033$ $= -0.033$	0.004356π
2.		0.5π	$\frac{\pi D^4}{128}$ $= \frac{\pi \times 2^4}{128}$ $= 0.125\pi$	$y_2 - \bar{y}$ $2 - 2.033$ $= -0.033$	0.0005445π
3.		0.25π	$0.055R^4$ $= 0.055 \times 1^4$ $= 0.055$	$y_3 - \bar{y}$ $1.57 - 2.033$ $= -0.463$	~~0.21~~ 0.053592π

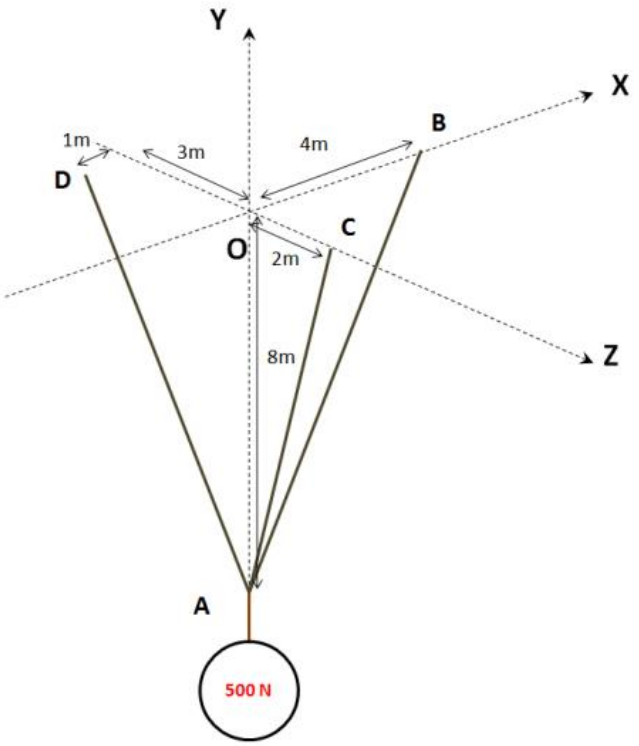
$\Sigma I_{Gxx} = 12.1186$ $\qquad \Sigma A(dy)^2 = -0.15639$

$$I_{xx} = \Sigma I_{Gxx} + \Sigma A(dy)^2$$

$$= 12.1186 + (-0.15639)$$

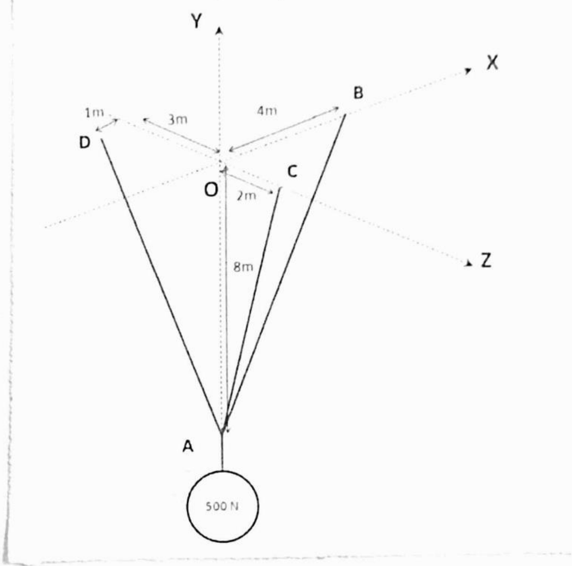
$$= \underline{11.96 \, m^4}$$

16	Three cables support a weight of 500 N at point A as shown in the figure. Determine the tension in the cables.

		Y X B 1m 3m 4m D C O 2m 8m Z A 500 N

Ans

Three cables support a weight of 500N at point A as shown in the figure. Determine tension in the cables.



A (0, −8, 0)

B (4, 0, 0)

C (0, 0, 2)

D (−1, 0, −3)

Next write the vector form of forces

$$\vec{T}_{AB} = |\vec{T}_{AB}| \cdot \frac{(4i + 8j + 0k)}{\sqrt{4^2 + 8^2 + 0^2}} = |\vec{T}_{AB}| \left(0.447i + 0.894j\right)$$

$$\vec{T}_{AC} = |\vec{T}_{AC}| \cdot \frac{(0i + 8j + 2k)}{\sqrt{0^2 + 8^2 + 2^2}} = |\vec{T}_{AC}| \cdot (0i + 0.970j + 0.242k)$$

$$\vec{T}_{AD} = |\vec{T}_{AD}| \cdot \frac{(-1i + 8j - 3k)}{\sqrt{(-1)^2 + 8^2 + (-3)^2}} = |\vec{T}_{AD}| \cdot (-0.116i + 0.929j - 0.348k)$$

Weight 500, $\vec{W} = 0i - 500j + 0k$

Since it is a case of equilibrium, if we find the resultant force at the point A will be zero. That means resultant vector will be zero vector.

$$\vec{R} = (0.447 + 0 + {}^{-}0.116)i + (0.894 + 0.970 + 0.929 + {}^{-}500)j + (0.242 + {}^{-}0.348)k = 0i + 0j + 0k$$

Resultant, R

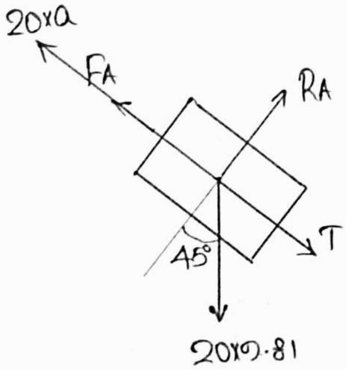
$$\Rightarrow (0.447|\vec{T}_{AB}| - 0.116|\vec{T}_{AD}|)i + (0.894|\vec{T}_{AB}| + 0.97|\vec{T}_{AC}| + 0.929|\vec{T}_{AD}| - 500)j + (0.242|\vec{T}_{AC}| - 0.348|\vec{T}_{AD}|)k$$

$$= 0i + 0j + 0k$$

		Now we can write $0.447\, T_{AB} - 0.116\, T_{AD} = 0$ $0.894\, T_{AB} + 0.97\, T_{AC} + 0.929\, T_{AD} = 500$ $0.242\, T_{AC} - 0.348\, T_{AD} = 0$ By solving these 3 equations, $T_{AB} = \underline{\underline{50.58\,N}}$ $T_{AC} = \underline{\underline{281.1\,N}}$ $T_{AD} = \underline{\underline{194.9\,N}}$
	17 a	Two masses MA = 20 kg and MB = 10 kg are connected by a bar of negligible mass. Find the acceleration of the system when it slides down an inclined plane of inclination 45° as shown in figure. Also find the force in bar. Assume μA= 0.2 and μB=0.4.

Ans

FBD of BLOCK A



$\Sigma F \perp r$ to plane $= 0$

$R_A - 20 \times 9.81 \cos 45 = 0$

$R_A = 138.73N$

$\cancel{\Sigma F \parallel \text{ to plane} = 0}$

$\cancel{T - F_A + 20 \times 9.81 \sin 45 = 0}$

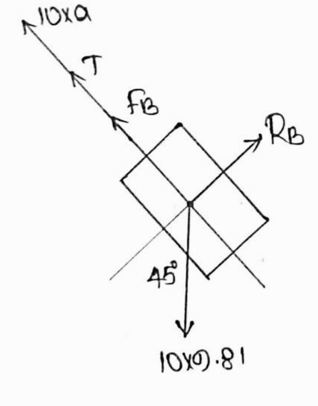
ΣF parallel to plane $= 0$

$T + (20 \times 9.81 \times \sin 45) - F_A - (20 \times a) = 0$

$T = 0.2 R_A + 20a - 20 \times 9.81 \times \sin 45$

$T = 20a - 110.98$ ——— (1)

FBD of BLOCK B



ΣF $\perp r$ to plane $= 0$

$R_B - 10 \times 9.81 \times \cos 45 = 0$

$R_B = 69.36\,N$

ΣF paralle to plane $= 0$

$10 \times a + T + F_B - 10 \times 9.81 \sin 45 = 0$

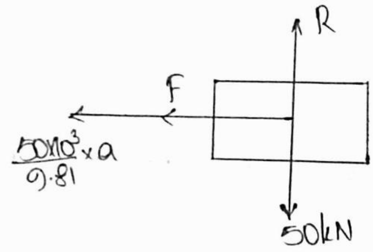
$T = 10 \times 9.81 \sin 45 - 10a - 0.4 \times 69.36$

$T = \cancel{70.36}\ 41.62 - 10a$ ——— (2)

(1) = (2)

$\Rightarrow 20a - 110.98 = \cancel{70.36}\ 41.62 - 10a$

		$30a = 152.60$ $a = \underline{\underline{5.08\ m/s^2}}$ $T = 20a - 110.98$ $= 20 \times 5.08 - 110.98$ $= \underline{\underline{-9.24N}}$ The negative sign indicates the force developed in the bar will be compressive force.
	17 b	A car moving at a speed of 60kmph, when the brakes are fully applies causing all four wheel to skid. Determine the time required to stop the car. The coefficient of friction between the road and tyre is 0.3. Weight of the car 50Kn

	Ans	 $\underline{\Sigma F_y = 0}$ $R - 50\times10^3 = 0$ $R = 50\times10^3 N$ $\underline{\Sigma F_x = 0}$ $-F - \frac{50\times10^3}{9.81}a = 0$ $-0.3\times50\times10^3 - \frac{50\times10^3}{9.81}a = 0$ $a = -0.3\times9.81$ $= -2.943 m/s^2$ Now, $v = 0$ $u = 16.66 m/s$ $a = -2.943 m/s^2$ $V = u + at$ $0 = 16.66 + -2.943\times t$ $t = \frac{16.66}{2.943} = 5.66$ sec
	18 a	A block of weight 50N is moving over a horizontal surface starting at rest, moves over a distance of 25m in 10 seconds under the action of a force of 20N. Determine the coefficient of friction between the surfaces.

Ans

Given: -

$W = 50N$

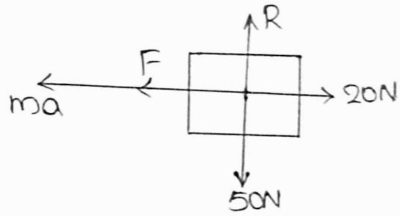
$u = 0$

$s = 25m$

$t = 10s$

$\mu = ?$

FBD of BLOCK



By applying D'Alembert's principle,

$\Sigma F_y = 0$

$R - 50 = 0 \Rightarrow R = 50N$ ——(1)

$\Sigma F_x = 0$

$20 - F - \frac{50}{9.81}a = 0$

$20 - \mu \times 50 - \frac{50}{9.81}a = 0$ ——(2)

		From given data, $S = ut + \frac{1}{2}at^2$ $25 = 0 + \frac{1}{2} \times a \times 10^2$ $25 = 50a$ $\frac{25}{50} = a$ $a = 0.5 m/s^2$ Substitute in equation (2) $\Rightarrow 20 - 50\mu - \frac{50 \times 0.5}{9.81} = 0$ $50\mu = 20 - 2.54$ $\mu = \frac{17.45}{50} = \underline{\underline{0.349}}$
	18 b	A car starts from rest on a curved road of radius 250 m and attains a speed of 18 km/hour at the end of 60 seconds while travelling with a uniform acceleration. Find the tangential and normal accelerations of the car 30 seconds after it started.

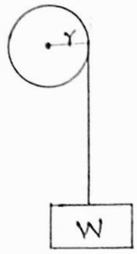
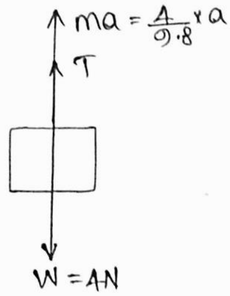
	Ans	Given:- $u = 0$ $r = 250m$ $v = 18 kmph = 18 \times \frac{5}{18} = 5 m/s$ $t = 60 sec$ We know, $v = u + at$ $5 = 0 + a \times 60$ $a = \frac{5}{60} = \underline{\underline{0.0833 m/s^2}}$ ∴ Tangential acceleration, $a_t = \underline{\underline{0.0833 m/s^2}}$ Now we have to find the normal acceleration at $t = 30 sec$ after it started. $a_n = \frac{v^2}{r}$

		To find normal acceleration, firstly find the velocity after 30sec, $v = u + at$ $= 0 + 0.0833 \times 30$ $= 2.5 \, m/s$ Now, Normal acceleration, $a_n = \frac{v^2}{r}$ $a_n = \frac{2.5^2}{250} = \underline{\underline{0.025 \, m/s^2}}$ The tangential acceleration is a constant value, it will not change with time as given in the question.
	19 a	A particle moving with simple harmonic motion has velocities of 8 m/s and 4 m/s when at the distance of 1 m and 2 m from the mean position. Determine (i) amplitude, (ii) period, (iii) maximum velocity, and (iv) maximum acceleration of the particle.

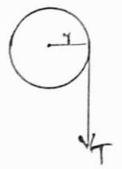
	Ans	
		Given:- $V_1 = 8 m/s$ $V_2 = 4 m/s$ $x_1 = 1m$ $x_2 = 2m$ $V = \omega\sqrt{r^2 - x^2}$ $8 = \omega\sqrt{r^2 - 1^2}$ ——— (1) $4 = \omega\sqrt{r^2 - 2^2}$ ——— (2) $\frac{(1)}{(2)} = \frac{8}{4} = \frac{\sqrt{r^2 - 1}}{\sqrt{r^2 - 4}}$ $\Rightarrow \frac{64}{16} = \frac{r^2 - 1}{r^2 - 4} \Rightarrow 4 = \frac{r^2 - 1}{r^2 - 4}$ $\Rightarrow 4r^2 - 16 = r^2 - 1$ $3r^2 = 15$ $r^2 = 5$ $r = \underline{2.23 m}$

		Subsitite in eq (1) $\Rightarrow 8 = \omega\sqrt{5-1}$ $8 = \omega \times 2$ $\omega = \underline{\underline{4 \text{ rad/s}}}$ ii) Time period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = \underline{\underline{1.57 \text{ sec}}}$ iii) Maximum velocity, $V_{max} = \omega \cdot r$ $= 4 \times 2.23 = \underline{\underline{8.92 \text{ m/s}}}$ iv) Maximum Acceleration, $a_{max} = -\omega^2 \cdot r$ $= -4^2 \times (2.23)$ $= -16 \times 2.23$ $= -35.68 \ \cancel{\text{rad/s}^2} \text{ m/s}^2$ $a_{max} = \underline{\underline{35.68 \text{ m/s}^2}}$
	19 b	A weight of 50 N suspended from a spring vibrates vertically with an amplitude of 7.5cm and a frequency of 1oscillation /second. Find the stiffness of the spring and the maximum tension induced in the spring

	Ans	Given:- $W = 50N$ $r = 7.5cm = 7.5\times10^{-2} m$ $f = 1 Hz$ $T = \frac{1}{f} = \frac{1}{1} = 1 sec$ $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{W}{kg}}$ $1 = 2\pi\sqrt{\frac{50}{k\times9.81}}$ $1 = 4\pi^2\times\frac{50}{k\times9.81}$ $k = \underline{201.21 N/m}$ Maximum Tension, T_{max} $T_{max} = 201.21\times7.5\times10^{-2}$ $[F_{max} = k.a]$ $= \underline{15.09 N}$
	20 a	A weight of 4 N is suspended by a light rope wound round a pulley of weight 48 N and radius 25 cm, the other end of the rope being fixed to the periphery of the pulley. If the weight is moving downwards, determine: (i) Acceleration of the weight 4 N, and (ii) Tension in the string. Take g = 9.80 m/s².

	Ans	<u>Given:-</u> $W = 4N$ $W_P = 48N$ $r = 25cm = 25 \times 10^{-2} m$  <u>FBD of BLOCK</u>  $T = 4 - \frac{4}{9.8} \times a$ —— (1)

FBD of pulley



Net moment = Force x radius

Torque = Force x radius

$\tau = I\alpha$ = Tension x radius

$$I = \frac{mr^2}{2} = \frac{\frac{4.8}{9.8} \times (25\times10^{-2})^2}{2}$$

$$\alpha = \frac{a}{r} = \frac{a}{25\times10^{-2}}$$

Now, $I\alpha = T\times r$

$$\frac{24}{9.8} \times (25\times10^{-2})^2 \times \frac{a}{25\times10^{-2}} = T \times 25\times10^{-2}$$

$$T = \frac{24a}{9.8} \quad \text{------} (2)$$

$$(1) = (2) \Rightarrow \frac{24a}{9.8} = 4 - \frac{4}{9.8}a$$

$$\frac{\overset{28}{\cancel{20}}a}{9.8} = 4$$

$$a = \frac{4\times9.8}{\cancel{20}\,28} = \cancel{1.96\,m/s^2} \quad \underline{1.4\,m/s^2}$$

		Substitute in eq (2) $\Rightarrow T = \dfrac{24 \times 1.4}{9.8} = \dfrac{24 \times 1.4}{9.8}$ $T = \underline{\underline{3.428N}}$
	20 b	A wheel, rotating about a fixed axis at 30 r.p.m. is uniformly accelerated for 50 seconds, during which time it makes 40 revolution. Find: (i) angular velocity the end of this interval, and (ii) time required for the speed to reach 80 revolution per minute.

Given:-

$N_0 = 30 rpm$

$\omega_0 = \frac{2\pi \times 30}{60} = \pi \; rad/s$

$t = 50 \; sec$

$\theta = 40 \times 2\pi = 80\pi \; rad$

$\omega = ?$

$\omega = \omega_0 + \alpha t$

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

$\omega^2 = \omega_0^2 + 2\alpha\theta$

By using ~~third~~ second equation,

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

$80\pi = \pi \times 50 + \frac{1}{2}\alpha \times 50^2$

$$1250\alpha = 30\pi$$

$$\alpha = \frac{30\pi}{1250} = 0.024\pi \text{ rad/s}^2$$

$$\alpha = 0.075 \text{ rad/s}^2$$

By using first equation,

$$\omega = \omega_0 + \alpha t$$

$$\omega = \pi \cancel{\times 50} + 0.024\pi \times 50$$

$$= \cancel{51.2\pi}\ 2.2\pi$$

$$\omega = \underline{\underline{6.91 \text{ rad/s}}}$$

ii) Time required for speed to reach 80rpm

$$\omega = \frac{2\pi \times 80}{60} = 2.66\pi \text{ rad/s}$$

$$\omega_0 = \pi \text{ rad/s}$$

$$\alpha = 0.024\pi \text{ rad/s}^2$$

$$\omega = \omega_0 + \alpha t$$

$$2.66\pi = \pi + 0.024\pi \times t \Rightarrow t = \underline{\underline{69.16 \text{ sec}}}$$